

Autodesk® Scaleform®

Scaleform 3D

Scaleform 3.2

3D

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Autodesk® Scaleform® 4.4

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1

Scaleform® 3Di™

UI

3D Flash

Scaleform 3.2 ,Scaleform

ActionScript ,

3D

Scaleform 3.2 ActionScript 2.0 ,

3D AS --- AS 3.0

HUD (_z _zscale _xrotation _yrotation _matrix3d _perspfov),
UI 3D 3D

c + +

3D

Scaleform

API

API

3D Gfx::Value

, 3D 3D Flash ,

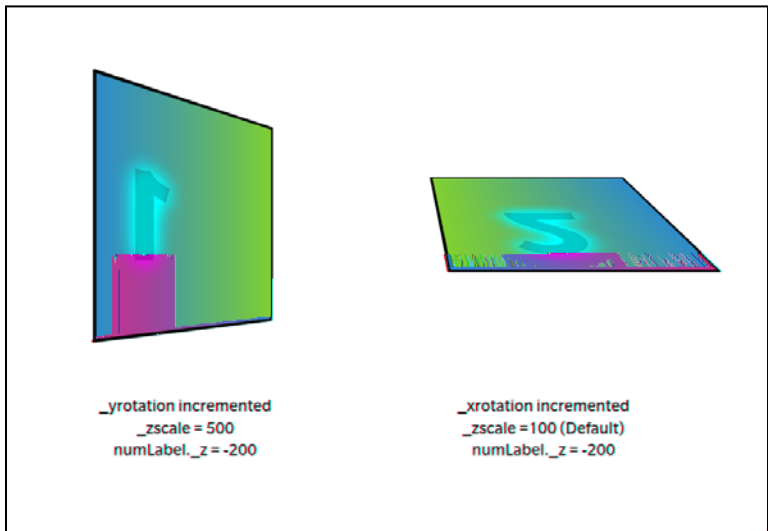
2 Flash 10

Scaleform 3Di Flash 10 3D , 3Di

2.1

Flash 10 3D , 3Di :

- 1. , , ,
- 2. (Back-face culling)



1 (Culled)

2.2

Scaleform 3Di , Flash 10 3D ,

2D 3D , 3D 3D
Flash 10

3 3D Flash

Scaleform 3.2, 3D Flash

3.1

Flash , 3D Flash
Scaleform 3.2 , 3D Flash
“(Render to Texture) 3Di Flash
3D , 3D , 3D
Z() / (UI)
, ,
Flash (Scaleform) ,
, Flash ,
, “SWF ”(SWF to Texture) , Scaleform SDK

3.2 3Di

(Scaleform 3.2) 3Di --- ActionScript Direct Access API (C++)
,3Di ,
— ,
Scaleform Scaleform
,3Di , Scaleform , UI,
, UI ,
UI 3D , UI ,
, UI ,

4 3Di API

4.1 ActionScript 2 API

3Di,Flash

```

    ,
    ,
    3D
    ,
    3D
    (x y z
    3d ),
    3D
    3d " "(Null)( , ActionScript ,foo._matrix3d = null),
    2D
    z 0 3D
    ,3D
    3 (x y z )
    ,
    Z 2D
    ,3D
    , _zscale ActionScript

```

Scaleform 3D Flash 10 +Z () +Z ,FOV 55 , +X ,+Y ,

3D Flash

-
-
-

3Di ActionScript

- **_z** - Z (), 0
 - **_zscale** - Z , 100
 - **_xrotation** - X() , 0
 - **_yrotation** - Y() , 0
 - **_matrix3d** - 16 (4 x 4) 3D
NULL, 3D , 2D
 - **perspfov** - (FOV) > 0 < 180
-1 FOV
- 55

	ActionScript	gfxExtensions	True()
1	<pre> _mc1._xrotation = 45; _mc1._yrotation = 45; _global.gfxExtensions = true; mc1._yrotation = 45; </pre>	<pre> _xrotation _yrotation </pre>	<pre> 3D 3Di </pre>
2	<pre> _sq1._xrotation = 45; _sq1._yrotation = 45; _global.gfxExtensions = true; sq1._zscale = 500; function rotateAboutXAxis(mc:MovieClip):Void { mc._xrotation = mc._xrotation + 1; if (mc._xrotation > 360) { mc._xrotation = 0; } } onEnterFrame = function() { rotateAboutXAxis(sq1); } </pre>	<pre> X Z </pre>	
3	<pre> function PixelsToTwips(iPixels):Number </pre>	<pre> 3D 3D </pre>	

```

{
    return iPixels*20;
}

function TranslationMatrix(tX, tY, tZ):Array
{
    var matrixC:Array = [ 1, 0, 0, 0,
                          0, 1, 0, 0,
                          0, 0, 1, 0,
                          tX,tY,tZ,1];
    return matrixC;
}

this._matrix3d = TranslationMatrix(PixelsToTwips(100),PixelsToTwips(100),0);

```

4.2 C++ 3Di

4.2.1 Matrix4x4

Matrix4x4 4x4 3D
() , Render_Matrix4x4.h

4.2.2 Direct Access

Direct Access (Gfx::Value) 3Di API 3D
3Di ActionScript

, Gfx_Player.h [Gfx::Value::DisplayInfo](#)

```
void SetZ(Double z)
void SetXRotation(Double degrees)
void SetYRotation(Double degrees)
void SetZScale(Double zscale)
void SetFOV(Double fov)
void SetProjectionMatrix3D(const Matrix4F *pmat)
void SetViewMatrix3D(const Matrix3F *pmat)
bool SetMatrix3D(const Render::Matrix3F& mat)
```

```
Double GetZ() const
Double GetXRotation() const
Double GetYRotation() const
Double GetZScale() const
Double GetFOV() const
const Matrix4F* GetProjectionMatrix3D() const
const Matrix3F* GetViewMatrix3D() const
bool GetMatrix3D(Render::Matrix3F* pmat) const
```

4.2.3 Render::TreeNode

Render::TreeNode , 3D “ ”(View) “
”(Perspective) , ,

Gfx_Player.h

```

void    SetProjectionMatrix3D(const Matrix4F& m)
void    SetViewMatrix3D(const Matrix3F& m);

```

4.2.4 : Direct Access API

```

Ptr<GFx::Movie> pMovie = ...;
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "_root.Window");
if (bOK)
{
    GFx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        // set perspectiveFOV to 150 degrees
        Double oldPersp = dinfo.GetFOV();
        dinfo.SetFOV(150);
        tmpVal.SetDisplayInfo(dinfo);
    }
}

```

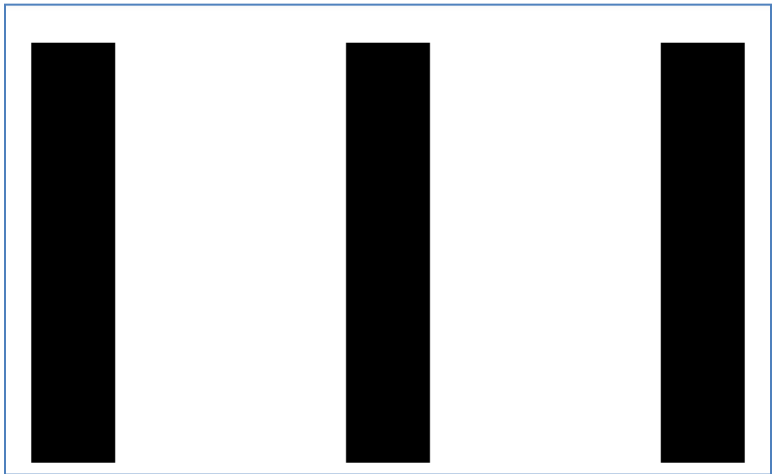
5

Direct Access API (FOV)

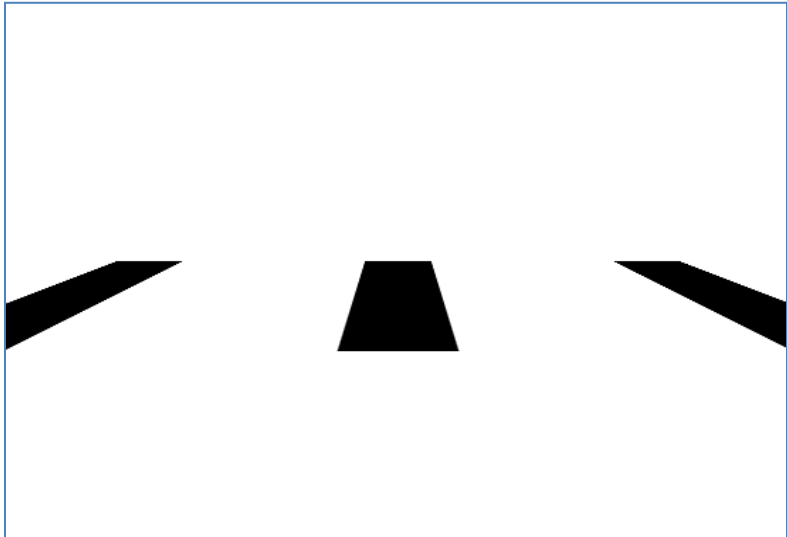
FOV 0, FOV

, z , FOV

0 180, 1, , 179 ,



2 Flash



3 -80 55 X

```

        FOV (5 )
        .
        ,
        .
        ActionScript property _perspfov -1 C++
        Direct Access API :

// 1. Perspective FOV -1

Gfx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root"); //"_root" for AS2
if (bOK)
{
    Gfx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        dinfo.SetPerspFOV(-1);
        tmpVal.SetDisplayInfo(dinfo);
    }
}

// 2.

void MakeOrthogProj(const RectF &visFrameRectInTwips, Matrix4F *matPersp)
{
    const float nearZ = 1;
    const float farZ = 100000;
    float DisplayHeight = fabsf(visFrameRectInTwips.Height());
    float DisplayWidth = fabsf(visFrameRectInTwips.Width());
    matPersp->OrthoRH(DisplayWidth, DisplayHeight, nearZ, farZ);
}
Gfx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root.mc");
if (bOK)
{
    Gfx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        Matrix4F perspMat;
        MakeOrthogProj(PixelsToTwips (pMovie->GetVisibleFrameRect()), &perspMat);
        dinfo.SetProjectionMatrix3D(&perspMat);
        tmpVal.SetDisplayInfo(dinfo);
    }
}
}

```

(Projection)

```
Ptr<Movie> pMovie = ...
// compute ortho matrix
Render::Matrix4F ortho;
const RectF &visFrameRectInTwips = PixelsToTwips(pMovie->GetVisibleFrameRect());
const float nearZ = 1;
const float farZ = 100000;
ortho.OrthoRH(fabs(visFrameRectInTwips.Width()), fabs(visFrameRectInTwips.Height()),
              nearZ, farZ);
pMovie->SetProjectionMatrix3D(ortho);
```

5.1 AS3

ActionScript 3 , PerspectiveProjection()

Example:

```
import flash.display.Sprite;

var par:Sprite = new Sprite();
par.graphics.beginFill(0xFFCC00);
par.graphics.drawRect(0, 0, 100, 100);
par.x = 100;
par.y = 100;
par.rotationX = 20;

addChild(par);

var chRed:Sprite = new Sprite();
chRed.graphics.beginFill(0xFF0000);
chRed.graphics.drawRect(0, 0, 100, 100);
chRed.x = 50;
chRed.y = 50;
chRed.z = 0;

par.addChild(chRed);

var pp3:PerspectiveProjection=new PerspectiveProjection();
pp3.fieldOfView=120;
par.transform.perspectiveProjection = pp3;
```

6 Stereoscopic 3D

Scaleform 3Di UI, 3D
, , , , ,
,

Scaleform 3Di 3D , ,PC NVIDIA 3D Vision Kit NVIDIA
, , , ()
, Display , ,
(Parallax Shift) Scaleform 4 , PS3
TinyPlayerStereo(Scaleform):
Apps \Samples \GfxPlayerTiny \GfxPlayerTinyPS3GcmStereo.cpp

6.1 NVIDIA 3D Vision

6.1.1 3D Vision

NVIDIA 3D Vision Kit 3Di UI, NVIDIA
3D 3D (Stereoscopic 3D
Test)

6.1.2

Scaleform Player 3D Vision
Direct3D Scaleform Player, , Scaleform Player ,
command line argument -f NVIDIA ,
3D 3D , (CTRL + T)

6.1.3

, ,
, 3D Vision (Convergence)
,

3D Vision, IR, (CTRL + F3) /(CTRL + F4), Scaleform Player, (CTRL + F5) (CTRL + F6), "Stereoscopic 3D", "Set up stereoscopic 3D", "Set Keyboard Shortcuts" NVIDIA " "(Enable the advanced in-game settings) (CTRL + F5) (CTRL + F6) (CTRL + F7), (CTRL + F5) (CTRL + F6), 10 15 (CTRL + F5) (CTRL + F6), (CTRL + F7) NVIDIA API, 3D, 3D

API
<http://developer.nvidia.com/object/nvapi.html>.

http://developer.download.nvidia.com/presentations/2009/SIGGRAPH/3DVision_Develop_Design_Play_in_3D_Stereo.pdf.

6.2 Scaleform Stereo API

Scaleform 4.1 Render::HAL API, 3D 3D
HDMI 1.4

Scaleform 6.2.1

Scaleform 4

Render::Viewport::SetStereoViewport

SF4 (Viewport Stereo)

```
// ; ,  
  
View_Stereo_SplitV = 0x40,  
View_Stereo_SplitH = 0x80,  
View_Stereo_AnySplit = 0xc0,  
  
void SetStereoViewport(unsigned display);
```

6.2.1 Scaleform Stereo

```
Render::StereoParams s3DInfo;  
s3DInfo.DisplayDiagInches = 46; // 46 inch TV  
s3DInfo.DisplayAspectRatio = 16.f / 9.f;  
s3DInfo.Distortion = .75; // 0 to 1  
s3DInfo.EyeSeparationCm = 6.4; // this will default 6.4cm  
pRenderHAL->SetStereoParams(s3DInfo);
```

Display , Display
Scaleform :

```
pMovie->Advance(delta);  
pRenderer->GetHAL()->SetStereoDisplay(Render::StereoLeft);  
pMovie->Display();
```

```
pRenderer->GetHAL()->SetStereoDisplay(Render::StereoRight);  
pMovie->Display();
```

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3Di

Scaleform SDK

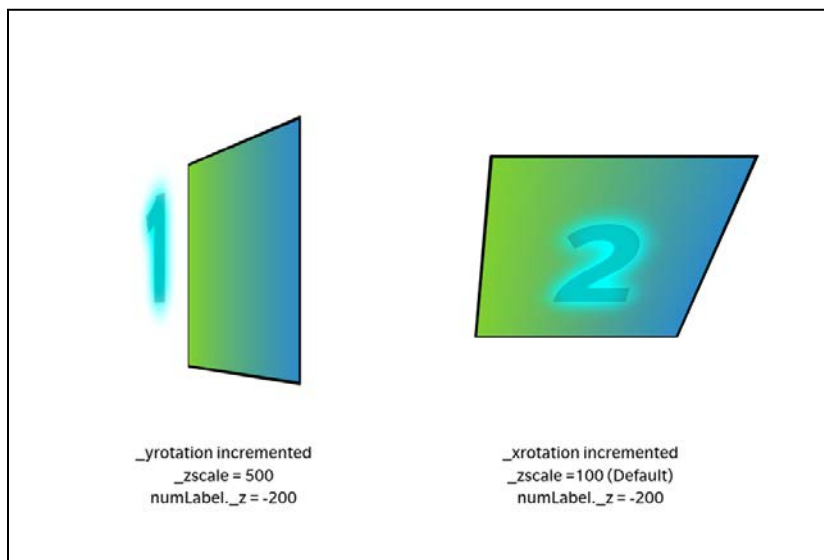
3D Flash

(SWF FLA)

,
C:\Program Files (x86)\Scaleform\GFx SDK 4.4\Bin\Data\AS2 or
AS3\Samples



4 3DGenerator



5 3D



6 3D